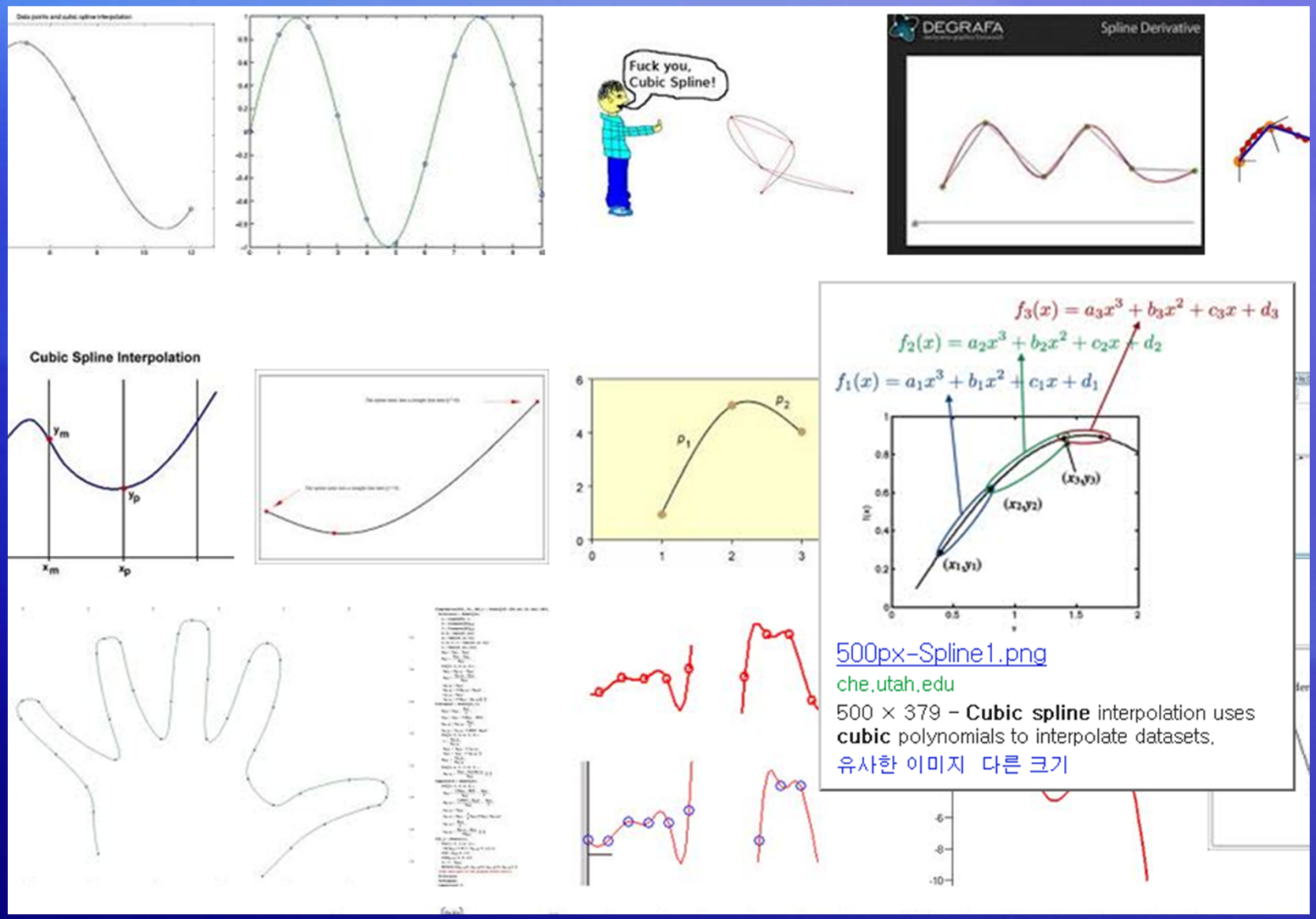
Google cubic spline

PS #5 - Q 1

In Do Hwang



Estimating productivity model

- *productivity = f (gender, dexterity, education)*

$$y = \alpha_0 + \alpha_1 sex + \alpha_2 dex + \alpha_3 lex + \alpha_4 lex^2 + u \quad , \quad sex = 1 \text{ if male}$$

	Coef.	Robust Std. Err.	P>t
sex	-0.901***	0.087	0.000
dex	0.112 ***	0.006	0.000
lex	0.821 **	0.321	0.011
lex2	-0.036 ***	0.013	0.005
cons	5.528 ***	2.032	0.007
obs=683	Prob>F=0.000	R2 = 0.3883	

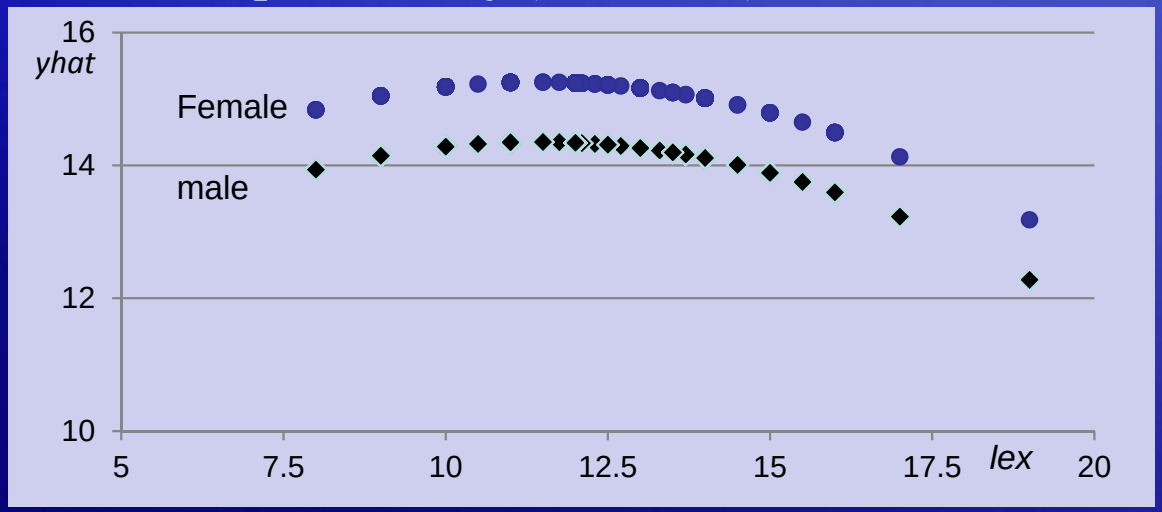
Test $H_0 : \alpha_4 = 0$

$H_0 : \alpha_3 = \alpha_4 = 0$

$F(2, 678) = 6.06$

Prob > F = 0.0025

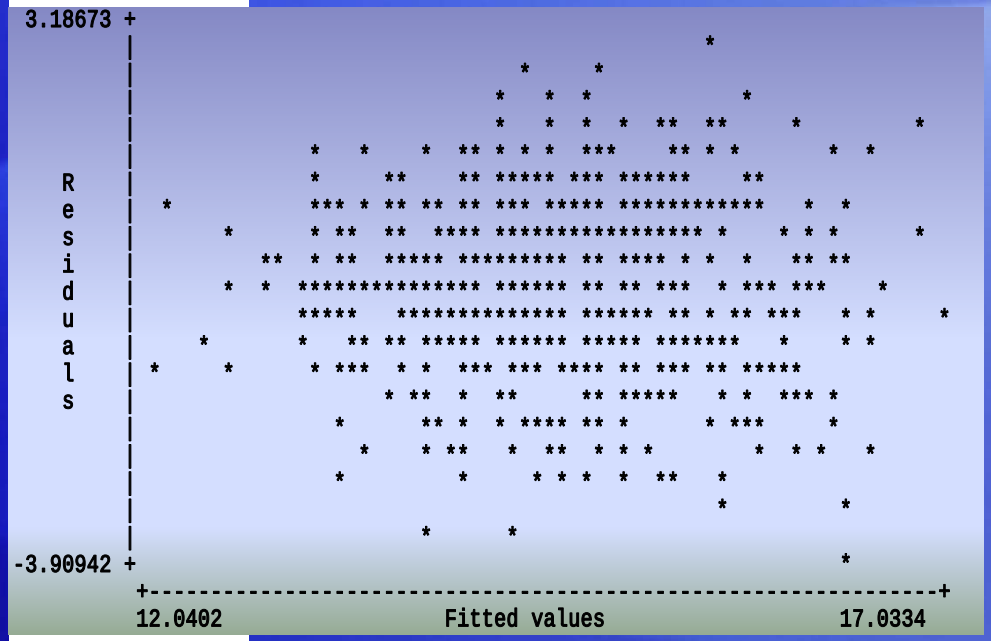
Estimated productivity (*dex*=44.95)

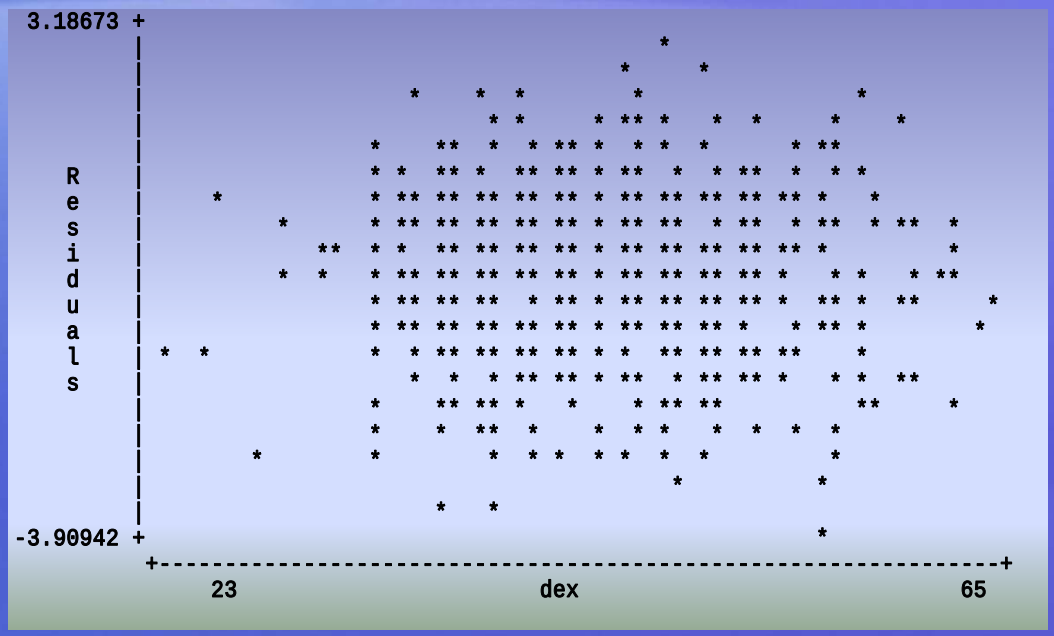


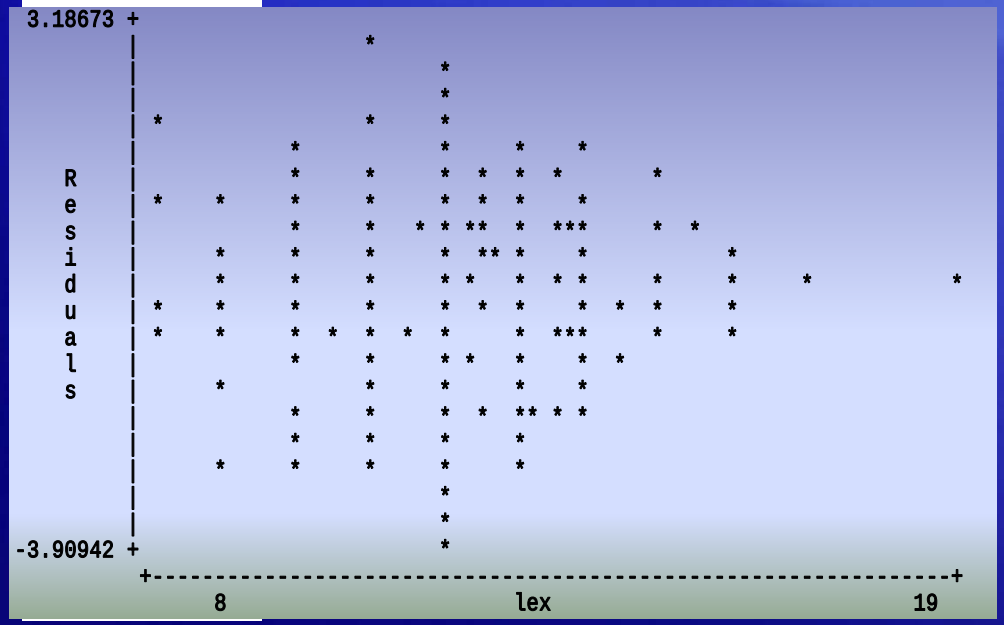
$$\hat{y} = -0.901 * sex + 0.112 * 44.95 + 0.821 * lex - 0.036 * lex2$$

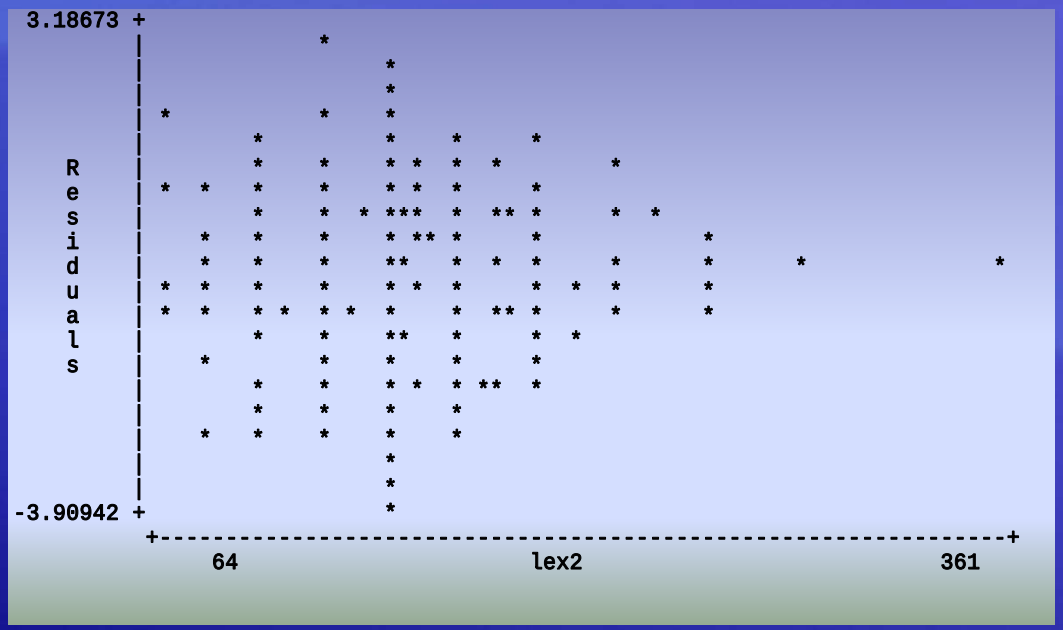
* sex =1 if male
=0 if female

Was that right model?









Was that right model?

- **Heteroskedasticity : Breusch-Pagan Test**

$$y = \alpha_0 + \alpha_1 x + u$$

$$u^{\wedge 2} = b_0 + b_1 x + v$$

➢ Breusch-Pagan test for heteroskedasticity

Ho: Constant variance

chi2(1) = 13.31

Prob > chi2 = 0.0003

- **Ramsey RESET Test**

$$y = \alpha_0 + \alpha_1 x + u \quad => \quad y^{\wedge}$$

$$y = b_0 + b_1 x + b_2 y^{\wedge 2} + b_3 y^{\wedge 3} + v$$

$$F = (R^2_{new} - R^2_{old})/q \div (1 - R^2_{new})/(n-b)$$

➢ Ramsey RESET test using powers of the fitted values of y

Ho: model has no omitted variables

F(3, 675) = 1.35

Prob > F = 0.2586

Was that right model?

If we seperately estimate…

$$y = b_1 + b_2\,dex + b_3\,lex + b_4 lex^2 + u \quad \ldots \text{ male}$$

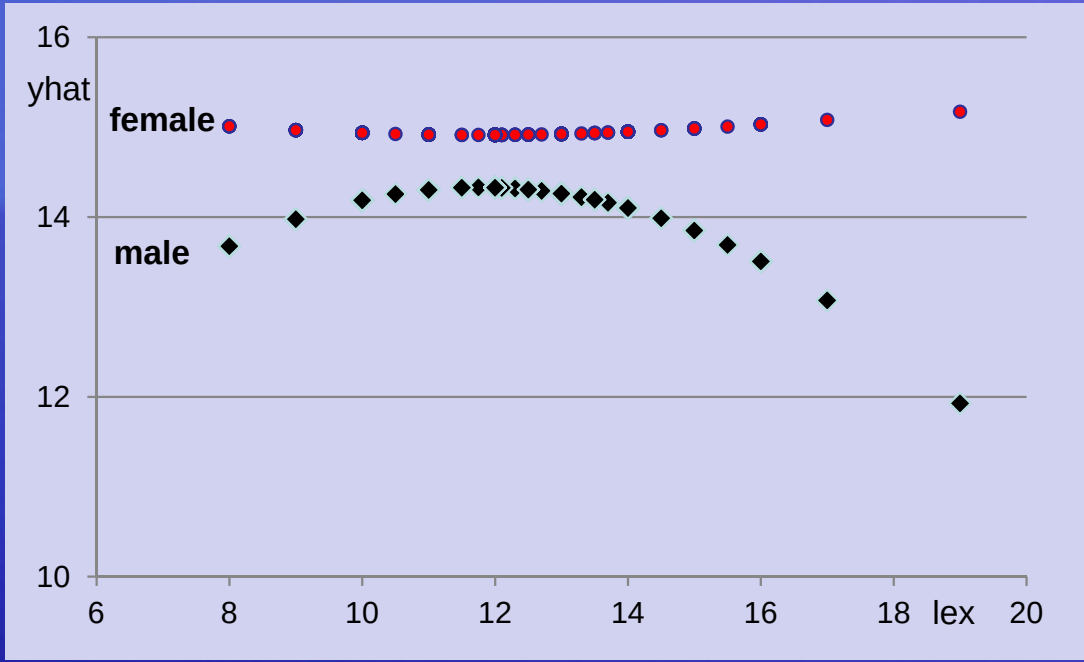
$$y = d_1 + d_2\,dex + d_3\,lex + d_4 lex^2 + v \quad \ldots \text{ female}$$

(cf) $y = \alpha_0 + \alpha_1 sex + \alpha_2 dex + \alpha_3 lex + \alpha_4 lex^2 + w$, $sex = 1$ if male

- **OLS result**

y	male	female
dex	0.107***	0.120***
(s.e.)	(0.007)	(0.011)
lex	1.083**	0.470
	(0.422)	(0.570)
lex2	-0.046 ***	-0.022
	(0.016)	(0.024)
cons	3.146	7.302**
	(2.764)	(3.423)
obs	387	296

- **Estimated productivity (*dex*=44.95)**



To check our dummy model

- $y = b_1 + b_2\,dex + b_3\,lex + b_4 lex^2 + \underline{\boldsymbol{b_5\,sex + b_6 sex\ dex + b_7 sex\ lex + b_8 sex\ lex^2}} + w$,

sex=1 if male …(1)

(cf) $y = b_1 + b_2\,dex + b_3\,lex + b_4 lex^2 + u$ . . . *male*

$y = d_1 + d_2\,dex + d_3\,lex + d_4 lex^2 + v$ . . . *female* …(2)

- **OLS result**

y	Model (1) : dummy model		Model (2) : female case	
	coef	Robust s.e.	Coef.	Robust s.e.
const	7.302**	2.858	7.302**	2.861
dex	0.120***	0.011	0.120***	0.011
lex	0.470	0.457	0.470	0.457
lex2	-0.022	0.018	-0.022	0.018
sex	-4.155	3.867		
sexdex	-0.013	0.013		
sexlex	0.613	0.593		
sexlex2	-0.024	0.023		

Test $H_0 : b_5 = b_6 = b_7 = b_8 = 0$

F(4, 675) = 28.78
Prob > F = 0.0000

Test $H_0 : b_6 = b_7 = b_8 = 0$

F(3, 675) = 0.71
Prob > F = 0.5473

Delta method and Bootstrap method

◆ ***Productivity maximizing education(lex*) ?***

$$\hat{y} = 5.528 - 0.901sex + 0.112dex + 0.821lex - 0.036lex^2$$

FOC) $\hat{lex}^* = -\hat{\alpha}_3/2\hat{\alpha}_4 = 11.4$ years

(1) Delta method

$Dh(\theta) = [0 \;\; 0 \;\; 0 \;\; -1/2\alpha_4 \;\; \alpha_3/2\alpha_4^2]$

$Dh(\hat{\theta})var(\hat{\theta})Dh(\hat{\theta}) = 0.409$

$\Rightarrow$ 95% CI = [10.17, 12.65]

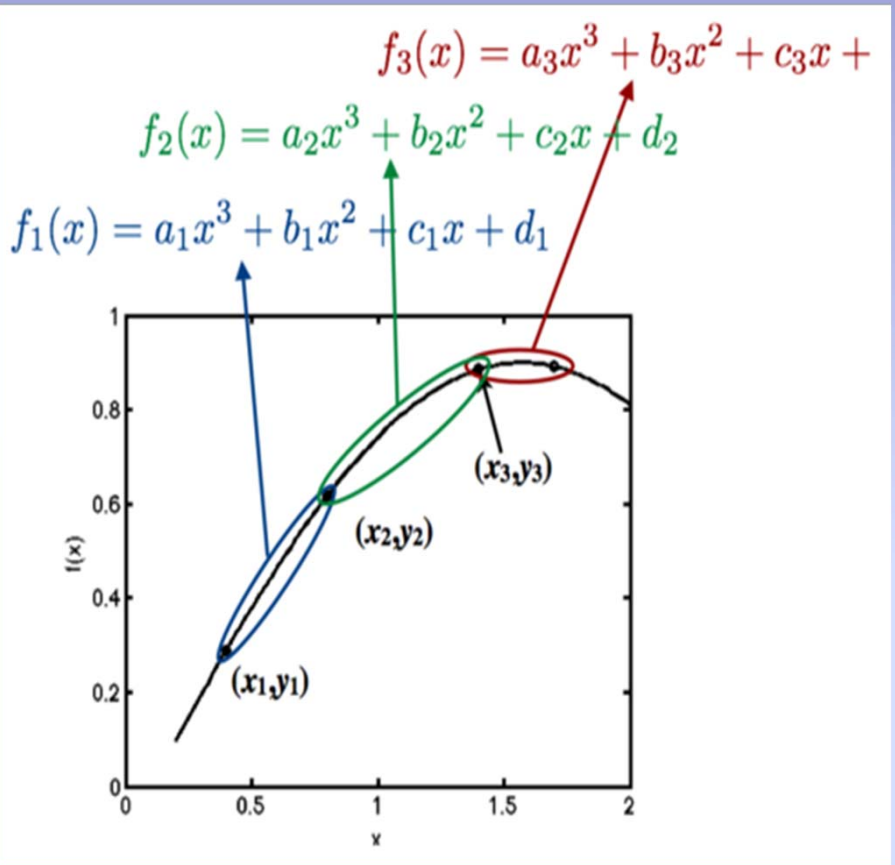
(2) Bootstrap

$\Rightarrow$ 95% CI = [8.5, 12.3]

• Bootstrap result

# of Replications	Observed	[95% conf. interval]	
30	11.3982	6.859	11.862
50	11.3982	6.611	12.283
100	11.3982	8.886	27.226
200	11.3982	9.636	12.293
500	11.3982	8.887	12.299
1000	11.3982	8.154	12.267
2000	11.3982	8.768	12.334
5000	11.3982	8.412	12.319
10000	11.3982	8.351	12.325
20000	11.3982	8.638	12.347

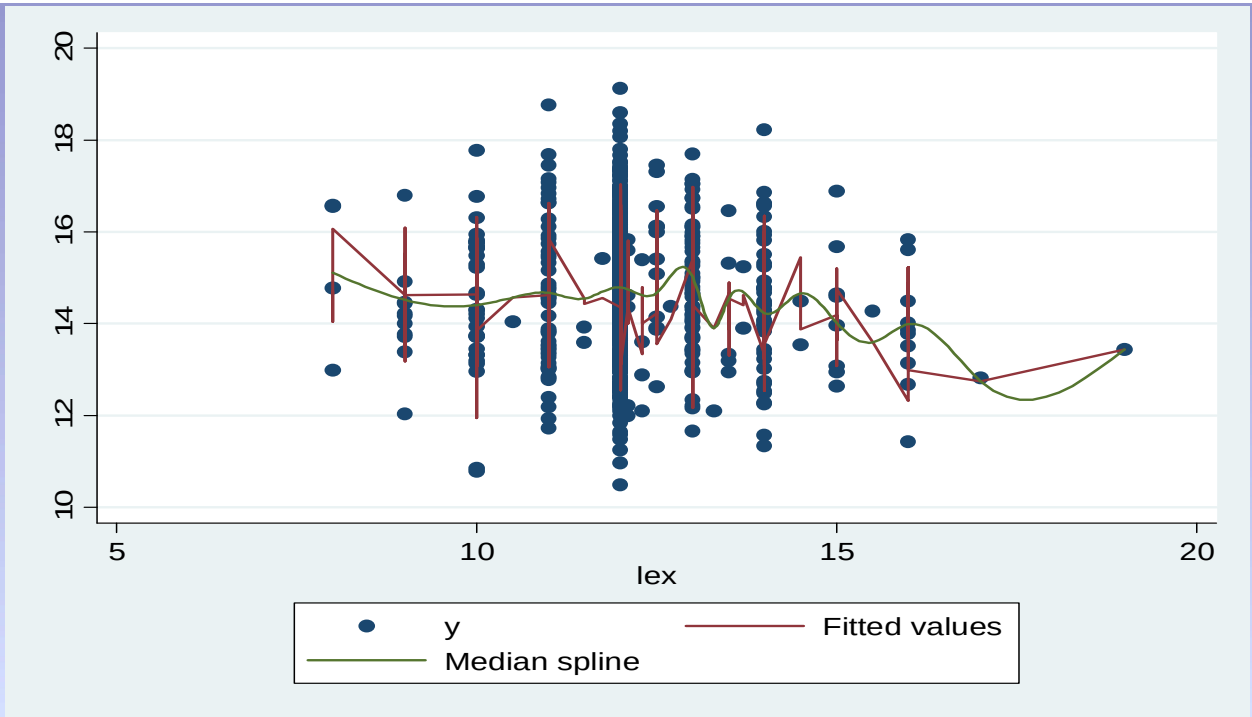
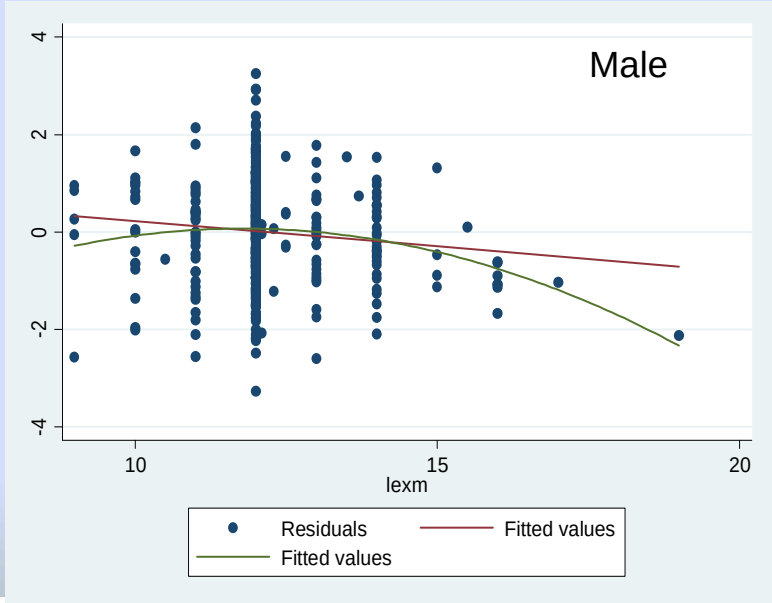
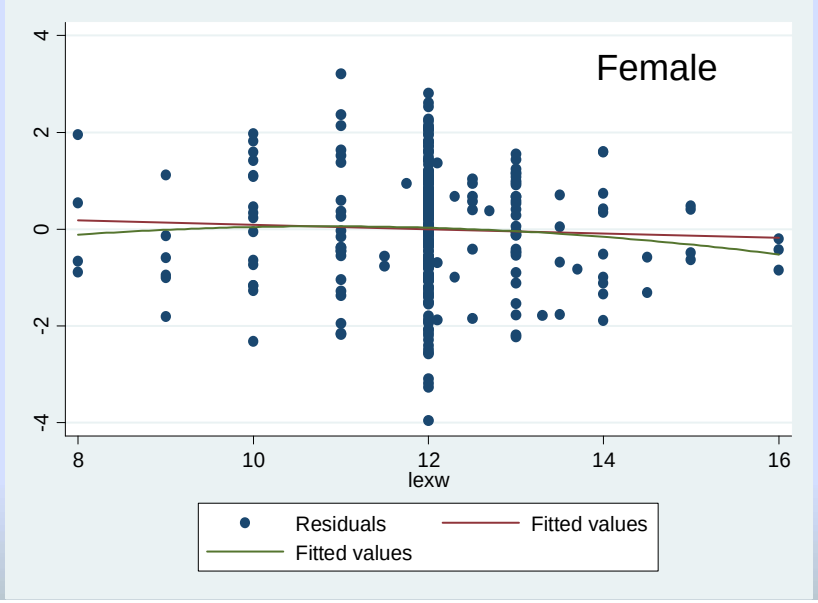
Cubic spline curve




```
. bspline, x(lex) kn(8(4) 20) power(3) gen(aa)
. line aa1 aa2 aa3 aa4 aa5 aa6 lex, sort
. regress y sex dex aa*,noconst
```

Source	SS	df	MS
Model	147559.304	8	18444.913
Residual	854.451683	675	1.26585435
Total	148413.756	683	217.296861

Number of obs	= 683
F(8, 675)	=14571.12
Prob > F	= 0.0000
R-squared	= 0.9942
Adj R-squared	= 0.9942
Root MSE	= 1.1251

y	Coef.	Std. Err.	t	P>|t|	[95% Conf.	Interval]
sex	-.8928062	.0879503	-10.15	0.000	-1.065495	-.7201172
dex	.1120877	.0060279	18.59	0.000	.1002521	.1239234
aa1	13.34161	4.84372	2.75	0.006	3.831035	22.85218
aa2	9.37088	.7778926	12.05	0.000	7.8435	10.89826
aa3	10.59231	.4503308	23.52	0.000	9.708088	11.47652
aa4	9.452501	.682117	13.86	0.000	8.113175	10.79183
aa5	7.973164	3.060423	2.61	0.009	1.964071	13.98226
aa6	.9223399	29.29183	0.03	0.975	-56.59172	58.4364

Cubic spline curve